

SOUTHEAST IOWA, IA, USA

WMO: 725420

Lat: 40.783N

Lon: 91.125W

Elev: 692

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) 2.4	(d) -10.7	(e) 3.2	(f) -1.8	(g) -6.2	(h) 4.1	(i) 3.3	(j) 24.9	(k) 23.6	(l) 22.5	(m) 24.2	(n) 8.0	(o) 310	(p) 0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.3	(c) 92.4	(d) 77.2	(e) 90.0	(f) 76.0	(g) 87.6	(h) 74.4	(i) 79.8	(j) 89.0	(k) 78.0	(l) 87.0	(m) 76.5	(n) 84.8	(o) 9.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	144.8	85.8	75.3	136.1	84.1	73.6	128.4	82.0	43.8	88.9	41.9	86.9	40.3	84.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.8	(b) 19.2	(c) 17.4	(e) -10.4	(f) 96.2	(g) 6.7	(h) 3.2	(i) -15.2	(j) 98.5	(k) -19.1	(l) 100.4	(m) -22.9	(n) 102.2	(o) -27.8	(p) 104.5		
(4) 21.8	(5) 19.2	(6) 17.4	(7) -11.0	(8) 82.6	(9) 6.5	(10) 1.5	(11) -15.6	(12) 83.7	(13) -19.4	(14) 84.6	(15) -23.1	(16) 85.4	(17) -27.8	(18) 86.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	24.4	29.0	40.8	52.9	63.4	72.7	76.0	74.2	67.4	55.0	41.8	30.3		
	DBStd	20.31	12.61	12.34	11.46	9.66	8.46	6.23	5.54	5.33	7.96	9.56	10.39	11.19		
	HDD50	2759	794	593	330	77	5	0	0	0	2	58	285	615		
	HDD65	5715	1259	1007	753	376	134	11	1	3	66	333	696	1076		
	CDD50	3654	0	6	45	165	421	681	805	751	523	214	39	4		
	CDD65	1137	0	0	2	14	86	243	341	289	137	24	1	0		
	CDH74	10227	0	0	18	147	715	2162	3335	2457	1201	189	3	0		
	CDH80	3598	0	0	0	24	192	763	1309	858	410	42	0	0		
(14) Wind	WSAvg	8.4	9.5	9.5	10.1	10.3	8.7	7.4	6.3	6.1	6.8	8.2	9.3	8.9		
(15) Precipitation	PrecAvg	34.6	1.6	1.1	2.9	3.8	3.6	3.9	3.7	3.5	4.1	2.6	2.0	1.7		
	PrecMax	50.1	3.9	2.0	5.9	7.4	7.0	9.0	7.2	8.1	11.7	7.2	3.5	5.0		
	PrecMin	25.9	0.2	0.2	0.8	0.7	1.1	0.9	1.2	0.6	0.0	0.1	0.2	0.1		
	PrecStd	6.5	1.1	0.6	1.3	2.0	1.6	2.0	1.6	2.1	3.1	1.7	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	64.8	77.3	83.4	89.8	93.1	97.5	94.3	92.4	85.4	72.6	62.7		
		MCWB	50.6	55.0	62.9	65.3	72.8	76.6	79.5	77.8	74.2	70.2	59.6	58.1		
	2%	DB	50.6	57.2	70.1	78.6	84.8	90.5	93.0	90.9	88.3	79.2	66.2	56.1		
		MCWB	45.3	50.0	58.8	62.7	70.1	75.6	78.5	77.0	72.5	65.5	56.6	50.8		
	5%	DB	45.3	51.6	63.8	73.8	81.5	87.8	89.9	88.0	84.8	74.5	62.0	50.1		
		MCWB	40.7	45.8	54.6	60.2	67.6	74.1	77.3	75.6	71.1	63.1	54.1	46.0		
	10%	DB	40.2	45.6	58.1	69.6	77.8	84.4	87.1	84.7	81.3	70.2	57.4	45.1		
		MCWB	36.8	41.1	50.1	57.6	65.6	72.5	75.3	73.7	69.1	61.3	50.7	41.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	57.5	64.6	68.2	75.3	79.9	82.6	81.3	77.2	72.2	62.6	59.2		
		MCDB	55.9	62.5	75.1	79.3	85.7	89.7	92.2	91.1	87.5	82.0	68.8	63.1		
	2%	WB	45.7	50.9	60.4	64.8	72.4	77.8	80.4	79.0	75.1	68.8	58.9	51.8		
		MCDB	50.3	56.7	67.8	74.6	82.0	87.8	89.6	87.8	84.6	75.4	64.1	54.4		
	5%	WB	40.8	45.8	56.2	62.2	70.1	76.1	78.6	77.3	73.2	65.7	55.5	46.0		
		MCDB	45.2	50.7	62.9	71.4	78.8	84.9	87.4	85.5	81.0	71.6	60.6	49.6		
	10%	WB	37.0	41.1	50.8	59.3	67.8	74.3	77.0	75.4	71.2	62.3	51.4	41.4		
		MCDB	40.0	45.8	57.4	67.8	75.3	82.4	85.2	82.7	78.7	69.3	56.9	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.9	18.1	20.0	19.1	18.5	18.3	18.3	20.9	19.5	16.9	14.3		
		MCDBR	20.8	23.2	24.6	25.9	22.3	20.8	20.1	20.9	23.5	24.2	22.4	19.5		
		MCWBR	16.6	16.6	15.4	14.6	11.6	10.1	9.4	10.0	10.4	13.0	15.3	15.9		
	5% WB	MCDBR	19.5	21.0	22.5	22.5	19.8	18.8	18.3	19.7	19.2	20.1	18.5			
		MCWBR	16.2	16.3	15.2	14.5	11.6	10.1	9.9	10.1	9.8	12.8	15.8	16.1		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.310	0.340	0.388	0.422	0.426	0.443	0.431	0.395	0.350	0.303	0.290		
	taud		2.453	2.395	2.400	2.267	2.234	2.281	2.205	2.254	2.310	2.438	2.511	2.465		
	Ebn at Noon		278	287	289	280	272	271	264	264	266	268	271	268		
	Edh at Noon		24	30	34	41	44	42	45	42	37	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		580	884	1180	1507	1762	1973	1972	1734	1485	992	665	491		
	RadStd		68	77	85	121	126	181	124	100	110	110	54	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page